hwsynth: Natural Synthetic Handwriting in 151 Unicode Scripts from Writer, Movement and Tool Models

pavel9860
https://github.com/pavel9860

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Abstract

Document understanding systems need handwriting in every script they meet, yet learned handwriting generators exist for a handful of scripts and need large per-script training sets, while rendering handwriting fonts looks mechanical: every *a* is the same *a*. We present `hwsynth`, a training-free generator that writes any text in 151 of the 175 Unicode scripts. Letters are traced from handwriting fonts (Noto where none exists) into pen paths; a writer model with 43 genes rewrites them; a movement model re-synthesises every stroke as overlapping lognormal submovements and trades shape fidelity against jerk energy by an effort parameter; a tool model with 17 writing instruments deposits ink by nib shape, ink rheology, speed and pressure. Twenty handwriting types, grounded in the discriminating elements of forensic handwriting examination, can each be written in ten situational styles and with either hand. A novelty-weighted evolution, scored against real scanned handwriting with document-disjoint fit and test sets, fills a bank of realistic individuals per type while keeping the types apart in two writer-identification spaces. On held-out real handwriting the evolved bank reaches a Fréchet embedding distance of 0.245 against 0.453 for font rendering and 0.065 between two halves of the real data, and its twenty types are identified with 52 % top-1 accuracy (chance 5 %).

1 Introduction

Handwriting is everywhere in the documents that machines are asked to read: filled forms, letters, notes, prescriptions, archives. Training and testing text recognition, layout analysis and document dewarping on handwriting needs large amounts of it, with exact labels, in every script the system will meet. Collecting and annotating real handwriting at that scale is slow and, for most of the world's scripts, practically impossible.

Synthetic handwriting is the usual answer, and the field has largely converged on learned generators: recurrent networks for pen trajectories [10], GANs [8, 15], transformers [2, 23] and, most recently, diffusion models [21, 22]; see [5] for a survey. They produce convincing samples *for the scripts they were trained on*, almost always Latin and occasionally Arabic, Chinese or a few Indic scripts, because they need thousands of labelled lines per script and style. For the other scripts of Unicode, practitioners fall back to rendering handwriting fonts with random affine jitter. The result is recognisably mechanical: identical glyph instances, perfectly continuous strokes, uniform ink, straight baselines.

This paper asks whether a generator built from explicit models of *who* writes, *how the hand moves* and *what tool touches the paper* can close that gap without any trained network. The challenge is that each model must be faithful enough to remove the mechanical look, and their parameters must be set so that the output is both realistic and diverse, which no hand-tuning achieves across dozens of interacting parameters.

Contributions.

- A script-agnostic letter source: centrelines traced from handwriting fonts and Noto fallbacks, drawn per grapheme cluster or per shaped word, covering 151 Unicode scripts (§3.1).
- A writer model separating *identity* (forensic discriminating elements [14]) from *situation*, with 20 handwriting types × 10 styles × 2 hands (§3.2).
- A movement model combining lognormal submovements [24] with a minimum-jerk [7] trade-off controlled by effort, which orders the styles from medical scrawl to calligraphy (§3.3).
- A tool and ink model for 17 instruments after the ink-deposition classes of [9] (§3.4).
- A leakage-controlled evaluation and a joint evolution that scores realism against real handwriting and separation with image and pen-path writer-identification features [3, 17] (§3.5).

The code is released under CC BY-NC 4.0 (§6).

2 Related Work

Learned handwriting generation. Graves [10] generated online pen trajectories with an LSTM and mixture density outputs. Haines et al. [11] synthesised a specific person's handwriting by assembling and blending their glyphs. ScrabbleGAN [8] and GANwriting [15] generate offline word images conditioned on content and style; Handwriting Transformers [2] and VATr [23] condition on a few style samples; latent diffusion models [21, 22] currently give the most faithful styles. All require labelled handwriting in the target script and style, and none covers more than a few scripts.

Perturbation and font based synthesis. Before deep generators, synthetic training data came from geometric perturbation of real lines [31] and from fonts. Font rendering scales to any script with a font, but repeats glyphs exactly; our letter source keeps the coverage of fonts and removes the repetition by drawing every glyph as a fresh movement.

Models of handwriting movement. The Kinematic Theory [24, 25] represents a stroke as a sum of lognormal velocity impulses; its Sigma-Lognormal model reconstructs real trajectories [6] and has been used to design trajectories [1]. Minimum-jerk planning [7] explains the smoothness of point-to-point arm movements. We use both: lognormal impulses for timing and corner cutting, jerk energy for the effort trade-off.

Individuality of handwriting. Forensic examiners compare a set of discriminating elements: allograph design, connections, proportions, slant, spacing, baseline, commencements and terminations, line quality and natural variation [14]. Srihari et al. [29] quantified individuality statistically. Automatic writer identification uses contour-direction and contour-hinge distributions [3, 26] or path signatures [17], grounded in rough path theory [4, 19]. We use these features to measure and enforce that our types are distinguishable.

Ink and pencil. Franke and Rose [9] related pen force to ink deposition for solid, viscous and fluid inks; Sousa and Buchanan [28] modelled graphite on paper. Our tool model follows these classes at the cost of a renderer, not a physical simulation.

Evaluation and search. We report Fréchet distances [13] and k-NN precision and recall [16] on ImageNet ResNet18 [12] embeddings and on stroke features. The writer bank is evolved with a novelty term [18]; quality-diversity search [20] is a natural extension (§5).

text → scripts, bidi order, units → **letters** (font centrelines) → **writer** (identity + style genes: shape, layout, joins) → **hand** (lognormal sub-movements, minimum jerk, wrist pivots, tremor) → **tool** (width, density, texture) → **page** (paper, ruling, smudges, word boxes)

Figure 1: The generation pipeline. No stage is learned; the writer bank (§3.5) only selects genes.

3 Method

Figure 1 summarises the pipeline. Text is split into words and drawing units; each unit's glyph centreline is transformed by the writer, the strokes of a line are moved by the hand, and the tool deposits ink.

3.1 Letters in every script

Each character is assigned its Unicode script [30]. Scripts with independent letter forms (Latin, Cyrillic, Greek, Hebrew, Han, Hangul, …) are drawn per grapheme cluster, so every letter gets its own jitter; all others are shaped by HarfBuzz/Raqm and drawn per word, so joining and conjuncts come from the font. Right-to-left runs are reversed while digits and Latin runs keep their order. A writer uses its Latin handwriting font for any script that font covers, otherwise a handwriting font of the script (59 fonts with class labels cursive, print, brush), otherwise a Noto Sans font (158). A glyph is rasterised at 160 px, skeletonised, and traced into a graph whose free spurs are pruned and whose edges continue straight through junctions (smallest turning angle), giving ordered pen strokes. Glyph size is normalised per script by a reference letter (e.g. Han is written at twice the x-height). Of the 175 scripts in Unicode, 151 have at least one font covering 90 % of their core letters.

3.2 Writers: identity and situation

A writer is a vector of 43 genes in $[0, 1]$. Twenty-two are *identity* genes following the discriminating elements of [14]: size, lateral expansion, ascender and descender length, capital size, slant, letter and word spacing, line pitch, roundness and angularity (garland, arcade, angular and thread movement), joining, entry hooks and exit flicks, dot placement and form, pressure, the allograph set (font), the share of letters written in a second allograph, handedness, all-capital writing and zone flattening. The others are *situational*: jitter of size, slant, rotation, baseline and spacing (correlated between neighbouring letters, AR(1) with $\rho = 0.6$), speed, effort, decay towards word ends, slant drift, baseline wander, tremor, cross-outs, retracing and mixed capitals.

A *type* fixes a centre for every identity gene (twenty types, Table 4 in the appendix); an individual draws each centred gene within ±12 % of its range. A *style* remaps sit-

uational genes into style ranges, keeping each writer's relative position, and shifts a few identity genes (e.g. hurried writers join more and slant further). Left-handed writers (hand gene > 0.9, about 10 %) pull horizontal strokes from right to left, draw some loops clockwise, write more upright, hold broad nibs steeper and smudge wet ink.

Natural variation comes from four sources: a second allograph for a fixed share of letters, AR(1) letter jitter, stroke ends that overshoot or stop short, and per-stroke ink-flow drift. Rare events (cross-outs of a misspelt word, retraced strokes) are drawn from skewed priors.

3.3 Hand movement

Each stroke is reduced to virtual targets (Douglas–Peucker turning points). Segment j becomes a submovement with velocity $\mathbf{D}_j\Lambda(t; t_{0j}, \mu_j, \sigma_j)$, where Λ is the lognormal impulse response of the Kinematic Theory [24]; start times are spaced by $2\,o\,e^{\mu-\sigma^2}$, twice the overlap ratio o times the time of peak speed (isochrony). Fluent writers have small o and cut corners; discrete-stroke scripts (Han, kana, Hangul, Devanagari) get larger o. The integrated path is then the minimiser of

$$\min_{\mathbf{x}} \ \lambda(\epsilon)\,\|\mathbf{x}-\mathbf{p}\|^2+\|D_3\mathbf{x}\|^2, \qquad \log_{10}\lambda = -6+6.5\,\epsilon, \ (1)$$

with D_3 the third difference (jerk) and end points held: a writer with low effort ϵ follows the minimum-jerk path and lets the letter shape go, a writer with high effort keeps the shape and pays in jerk energy. Effort is a situational gene whose style ranges increase from medical (0–0.12) through hurried, professional, casual, educated and formal to accurate and calligraphic (0.9–1). The speed of the synthesised movement drives the tool. Lines follow wrist-pivot arcs that reset at hand shifts every 25–60 mm, plus slow random baseline wander.

3.4 Tools and ink

Seventeen tools combine a nib shape (round; broad edge with thick–thin contrast $w + e|\sin(\phi+\alpha)|$ for direction ϕ and nib angle α; brush with width growing as pressure$^{1.5}$ and tapered ends) with an ink class [9]: *solid* (graphite: density follows force, paper grain), *viscous* (ballpoint paste: density falls with speed, skips on fast strokes, a blob where the pen lands) and *fluid* (gel, rollerball, fineliner, fountain, markers: near-constant density, pooling at slow points, bleeding halo, darker overlaps). Pressure combines the writer's push, its variation, extra force on downstrokes, and lift-off ramps. Strokes are drawn as dense anti-aliased discs at 3× supersampling and combined with tool-specific overlap.

3.5 Writer bank: evolution with realism and separation

Real references. Text lines are cut from 980 scanned handwritten documents, cleaned (no rulings, staves or bars; contrast; ink share) and kept when most of their five nearest labelled lines are handwriting. The documents are split into a *fit* set (2 876 lines) used by the evolution and a *test* set (1 181 lines) used only for metrics; documents whose lines were used to label the handwriting filter are never test documents.

Fitness. An individual is rendered on four fitness texts, half in the casual style and half in a random other style, with two seeds. *Realism* is the mean of two z-scores: the mean cosine similarity of its line embeddings to their five nearest fit lines, and minus the mean distance of its stroke features to their five nearest fit lines, each standardised by the same statistic of fit lines against lines of other documents. *Separation* is the distance to the nearest other type's centroid minus the distance to the own centroid, in contour-direction/hinge space [3] (χ^2, three fragment lengths) and in pen-path signature space [17] (depth-3 signatures of fragments, z-scored), each normalised by its median own distance. *Novelty* is the mean embedding distance to the ten nearest individuals of the same type. All twenty types evolve jointly (fitness = realism +0.5 separation +0.5 novelty). An individual enters its type's bank only if a re-evaluation with fresh seeds also reaches the realism threshold, it is nearer its own type than any other, and it differs from every banked individual.

4 Experiments

Setup. Metrics use texts disjoint from the fitness texts and the held-out test documents. We compare: plain rendering of a handwriting font (*font*); writers with uniform random genes (*random*); type priors without evolution (*prior*); the evolved bank (*bank*); and the bank without the pen model, without shape variation, and without letter joins. Embeddings are ResNet18 (ImageNet) [12]; legibility is the character error rate of Tesseract [27] on English lines.

4.1 Realism and diversity

Table 1 compares the systems on the held-out test documents. Lower Fréchet distances (FD) and higher precision and recall mean closer to real handwriting; the first row is the reference level of real handwriting against other real handwriting.

4.2 Styles and effort

Table 2 lists the bank per style. Movement energy (jerk per unit length of the planned pen path) and turning entropy rise with the effort a style spends on letter shapes, from medical scrawl to calligraphy; legibility follows.

system	realism	FD emb.	FD feat.	precision	recall	diversity	writer id	CER	ms/line
real (other documents)	–	0.065	3.70	0.746	0.658	–	–	–	–
font rendering	-0.651	0.453	30.61	0.244	0.000	0.0991	0.244	0.078	5
random writers	0.356	0.250	20.24	0.688	0.064	0.1189	0.588	0.524	205
type priors	0.331	0.248	19.94	0.669	0.023	0.1322	0.544	0.403	158
evolved bank	0.383	0.245	19.13	0.688	0.015	0.1353	0.550	0.464	160
bank, no pen model	0.372	0.259	19.43	0.625	0.013	0.1325	0.475	0.455	150
bank, no shape variation	0.342	0.258	18.73	0.619	0.014	0.1376	0.544	0.430	149
bank, no joins	0.364	0.249	18.93	0.613	0.020	0.1340	0.475	0.388	151

Table 1: Systems against held-out real handwriting (casual style, texts disjoint from the fitness texts).

style	realism	FD emb.	energy	entropy	CER
medical	0.574	0.260	1.3e-07	0.882	0.690
hurried	0.528	0.254	2.7e-07	0.926	0.568
professional	0.391	0.276	6.8e-07	0.959	0.428
casual	0.383	0.245	1.3e-06	0.971	0.464
unsupported	0.504	0.254	1.2e-06	0.966	0.482
educated	0.384	0.269	3.3e-06	0.996	0.507
uneducated	0.452	0.284	1.5e-06	0.976	0.285
formal	0.269	0.284	5.9e-06	0.999	0.456
accurate	0.290	0.322	8.2e-06	0.997	0.209
calligraphic	0.280	0.252	3.1e-05	1.109	0.575

Table 2: Evolved bank per style, ordered by effort.

writers	features	type	individual	style
prior	contour hinge	0.367	0.302	0.446
prior	path signature	0.375	0.504	0.532
bank	contour hinge	0.485	0.306	0.444
bank	path signature	0.519	0.486	0.569
chance		0.050	0.0125	0.200

Table 3: Identification accuracy (20 types, 5 styles).

4.3 Distinguishable types, individuals and styles

Table 3 reports top-1 nearest-neighbour identification on held-out texts: a type must be recognised from *other* individuals of the type, an individual from its *other* texts, a style from *other* individuals.

Analysis. All model-based systems close about half of the gap between font rendering and real handwriting (embedding FD 0.245–0.259 against 0.453 for fonts and 0.065 for real handwriting of other documents), and they triple the precision. The remaining gap is large, and recall stays low (1–6 %). The real scans span archives, coloured paper, faded ink and scanner artefacts that clean synthetic lines on near-white paper do not cover, so recall measures the page model as much as the handwriting. Evolution adds less realism than the models do (0.383 against 0.331 for the type priors), but it makes the types much more distinguishable: type identification rises from 37 % to 49 % with contour-hinge features and from 38 % to 52 % with path signatures, against 5 % chance, while individuals stay identifiable (31–49 % against 1.25 %). Removing the pen model, the shape variation or the letter joins each lowers realism and precision. Movement energy rises strictly with the effort of the styles, from medical (1.3×10^{-7}) to calligraphic (3.1×10^{-5}), and the turning entropy rises with it. Accurate writing is the most legible of the styles for Tesseract (CER 0.21) and medical writing the least (0.69). Messy styles score higher realism than formal ones because the references are dominated by everyday handwriting. Generation takes about 150 ms per line on one CPU core.

5 Discussion and Limitations

The realism measure is a proxy: ImageNet embeddings and 22 stroke statistics see texture and layout, not the correctness of letterforms in a particular script, and the reference lines are mostly Latin. No trained writer identification network was available, so separation uses classical features; the deep writer embeddings of recent work would be a stronger test. Scripts without a handwriting font inherit Noto's letterforms, which our writer and movement models make handwritten in execution but not in allograph design. The evolution maximises per-individual fitness; quality-diversity search [20] over trait descriptors and fitting individuals to real documents by analysis-by-synthesis would cover the space of real writers more evenly.

6 Conclusion

Explicit models of the writer, the hand and the tool, set by a leakage-controlled evolution, produce natural, diverse handwriting in 151 scripts without any trained generator. Every layer is inspectable: one can ask for a left-handed elderly writer, in a hurry, with a pencil. Natural extensions are handwriting fonts for the remaining scripts, page-level phenomena (margins, insertions, marginalia), learned discriminators as realism judges, and fitting the bank to real writers. Code, examples and this paper: https://github.com/pavel9860.

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A Handwriting types and styles

type	fonts	description
school cursive	cursive	taught joined script: loops, right slant, garlands
round print	print	rounded upright print, open letters
compact print	print	small neat narrow print
block capitals	print	everything in capitals, drafting style
backhand	any	left-leaning script
angular cursive	cursive	sharp, narrow, strongly slanted joined script
thready	cursive	fast thread connections, letters dissolve
wide garland	cursive	wide open letters, generous spacing
narrow tall	any	narrow letters, long ascenders and descenders, tight spacing
micro	any	very small writing
expansive	any	large writing, big capitals, wide spacing
arcade	cursive	rounded tops, slow upright joined script
print script	any	print and script mixed
elderly	cursive	slow tremulous joined script
child	print	large print, uneven baseline, heavy pressure
left hooked	any	entry hooks, slight left slant, pressure on downstrokes
copperplate	cursive	steep slant, long loops, ornate capitals
sloppy fast	cursive	fluent but careless
ruled neat	any	very regular, even and aligned
artistic	cursive	embellished capitals, exaggerated loops and terminals
style	**tools**	**description**
casual	all	everyday notes: as the person writes
formal	fountain fine, fountain medium, ballpoint	letters, applications: careful and even
accurate	ballpoint, gel, fineliner	forms and labels: separated careful letters
calligraphic	broad nib, italic nib, brush pen	broad-nib pen calligraphy: slow, contrasted
professional	ballpoint, gel, rollerball	fast, fluent, consistent office notes
hurried	ballpoint, ballpoint fine, gel	written in a rush
medical	ballpoint, ballpoint fine, gel	prescription scrawl
educated	ballpoint, fountain fine, fountain medium	fluent practised writing
uneducated	ballpoint, pencil 2b, pencil hb	slow unpractised print, mixed capitals
unsupported	ballpoint, gel, pencil hb	on a sheet held in the hand, no support

Table 4: The 20 handwriting types and 10 styles; 17 tools.

B Examples

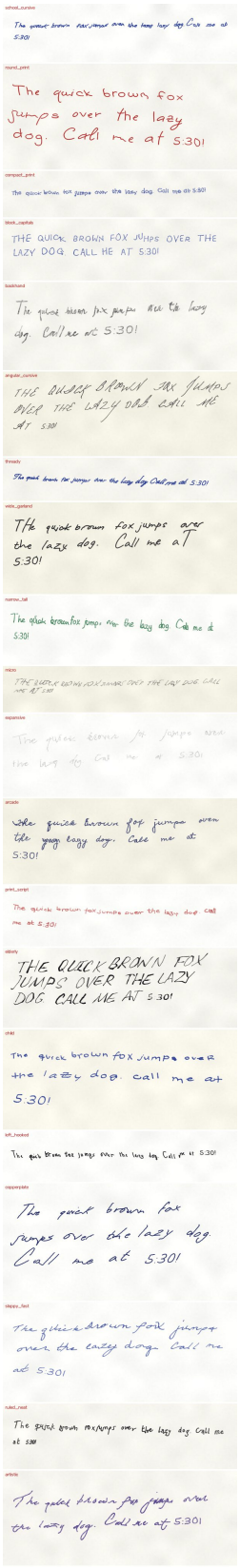


Figure 2: One individual of each of the 20 types (casual style).

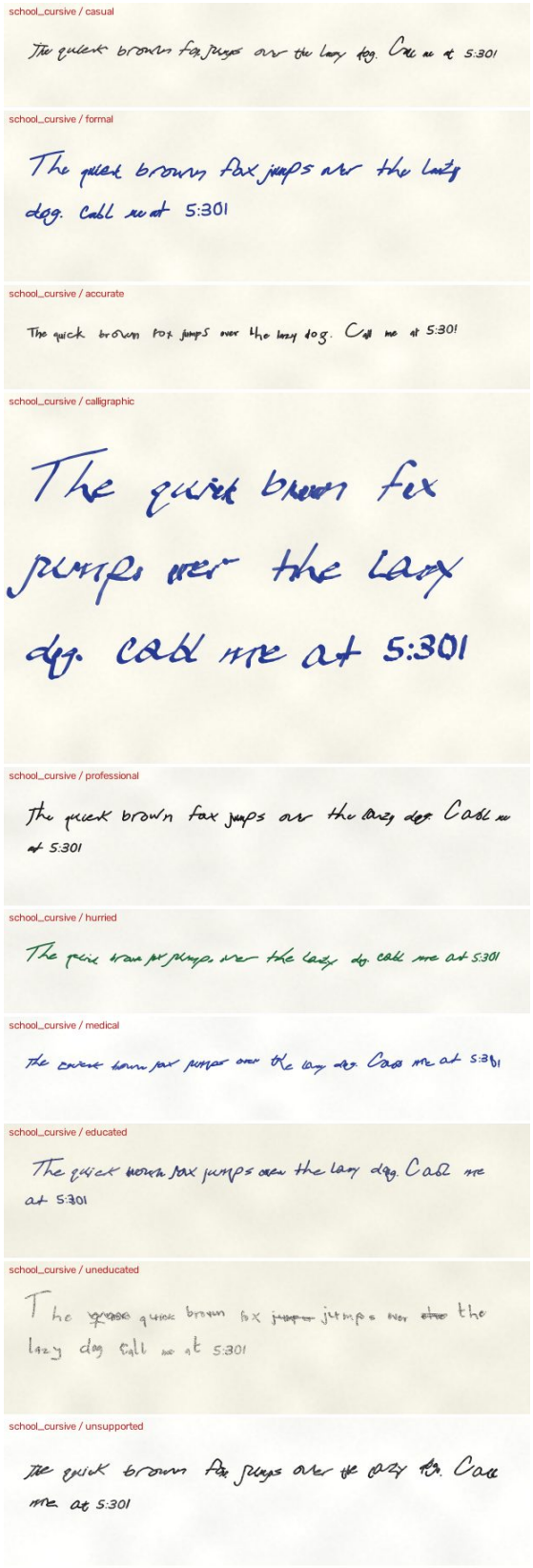


Figure 3: One school-cursive individual in the 10 styles.

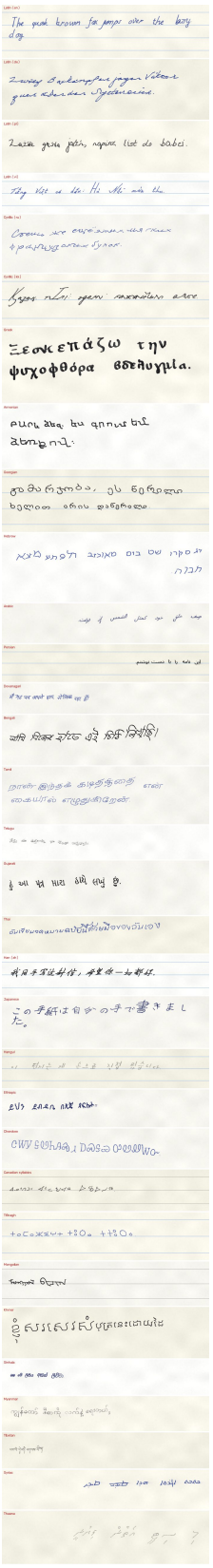


Figure 4: Random individuals writing 32 scripts.

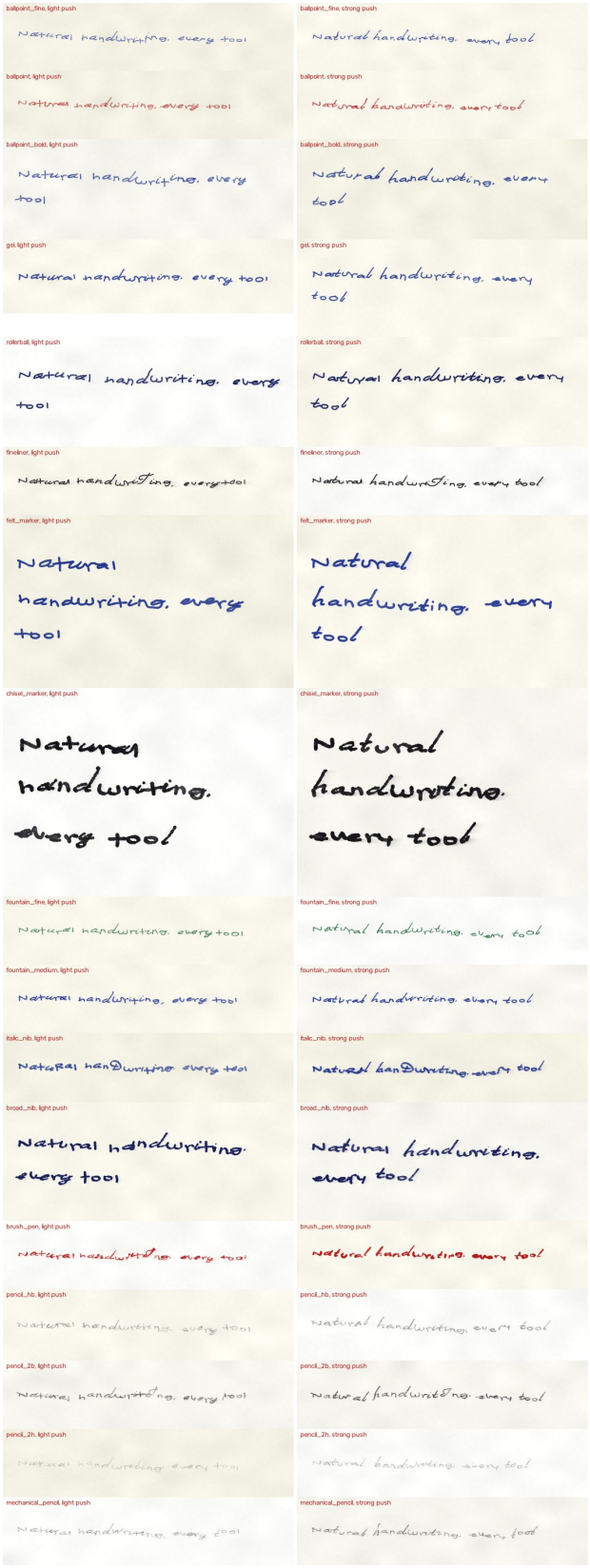


Figure 5: The 17 tools at light and strong push.

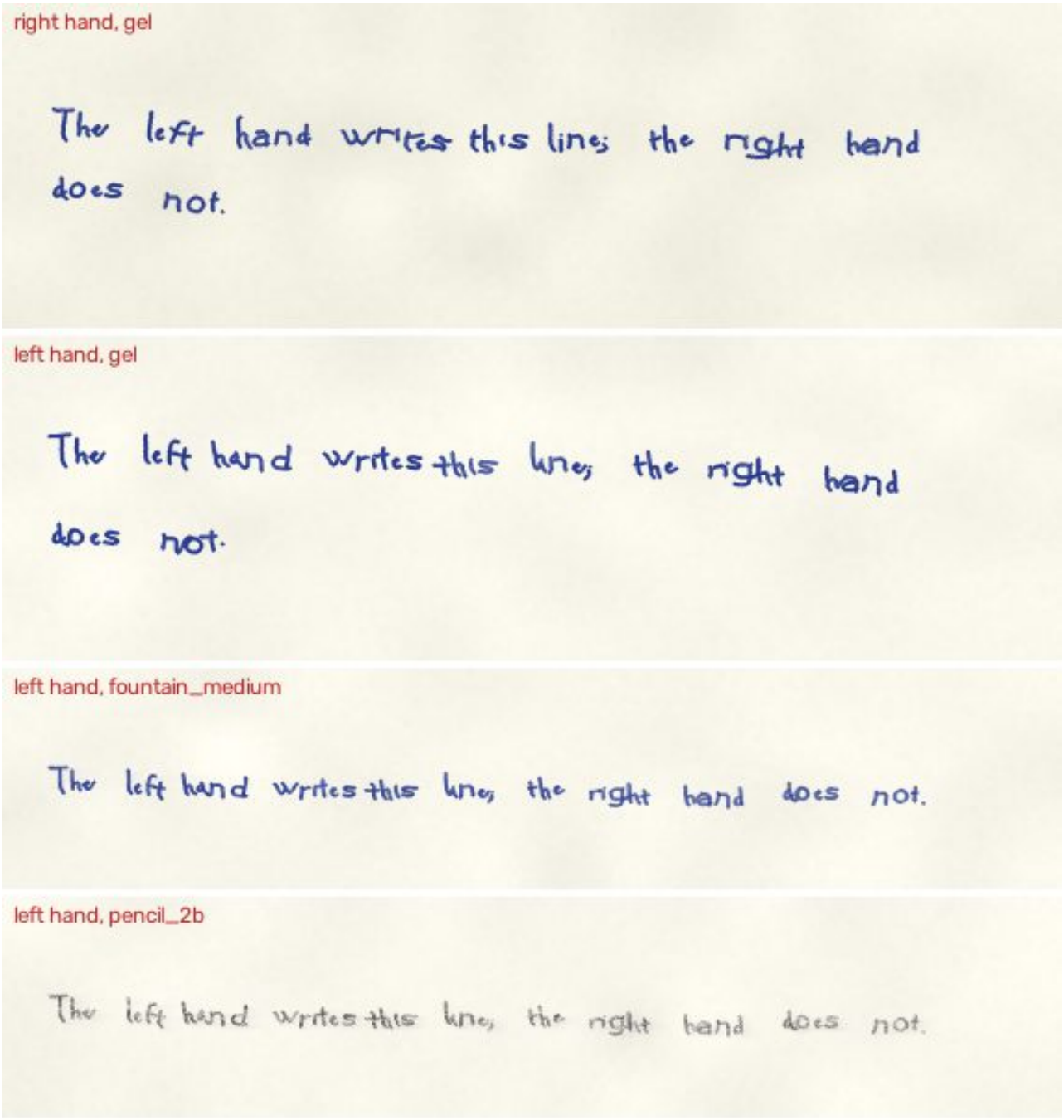


Figure 6: Right hand and left hand (gel, fountain pen, pencil): upright slant and smudges.